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Equine Endocrinology

Effects of the Sodium-Glucose Cotransporter-2 Inhibitor Velagliflozin on Insulin Concentrations in Horses With Insulin Dysregulation

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ABSTRACT

Background: Sodium-glucose cotransporter-2 inhibitors (SGLT2i) are promising treatments to manage hyperinsulinemia in horses with insulin dysregulation (ID).

Hypothesis: The SGLT2i velagliflozin decreases insulin concentration in horses with ID.

Animals: Privately-owned adult horses ($n = 37$) with laboratory-confirmed ID (low-dose oral sugar test insulin concentration > 75 μIU/mL).

Methods: Double-blind randomized placebo-controlled trial. Horses received placebo ($n = 19$) or velagliflozin 0.3 mg/kg PO q24h ($n = 18$) for 20 weeks (Study Period 1, SP1) immediately followed by a 20-week open-label trial where all animals received velagliflozin (SP2). Analysis of resting insulin, glucose, and triglyceride concentrations and body condition score (BCS) was performed between treatment groups and study periods using a Mann–Whitney *U* test. For SP1, analysis of changes in biochemical analytes over time was performed using generalized linear mixed effects models (GLMM). Data are reported as median (interquartile range).

Results: In SP1, GLMM indicated a significant effect of treatment on insulin concentration (71 [33–131] μIU/mL in horses receiving velagliflozin and 157 [82–298] μIU/mL in horses receiving placebo; $p < 0.0001$). The average (95% confidence interval [CI]) effect of velagliflozin treatment on insulin concentration was 155 (90–219) μIU/mL. Horses receiving placebo in SP1 had lower insulin (50 [26–99] μIU/mL) during SP2 ($p < 0.0001$). All horses experienced a transient increase in serum triglyceride concentration during velagliflozin treatment with no clinical abnormalities reported. In SP1, larger decreases in BCS occurred in horses receiving velagliflozin (median BCS 1 point lower than baseline; $p = 0.02$) than those receiving placebo.

Conclusions: Velagliflozin significantly decreased resting insulin concentrations in horses with ID.

Abbreviations: EDTA, ethylenediaminetetraacetic acid; EMS, equine metabolic syndrome; ID, insulin dysregulation; MOL, modified Obel lameness; OST, oral sugar test; PPID, pituitary pars intermedia dysfunction.

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1 | Introduction

Insulin dysregulation (ID) is a common endocrine disorder that substantially impacts the health of horses because of its association with laminitis [1]. Affected animals exhibit derangements in insulin status, including resting or postprandial hyperinsulinemia, tissue insulin resistance, or a combination of these abnormalities. Insulin dysregulation is the central component of equine metabolic syndrome (EMS), a collection of risk factors for laminitis that includes a predisposition toward obesity, regional adiposity, and altered blood adipokine concentrations [2, 3]. Hyperinsulinemia has been shown to lead to laminitis in horses [4, 5], and the risk of laminitis increases with higher blood insulin concentrations [6–8].

The primary goal of managing EMS is to lower blood insulin concentrations to minimize the risk of laminitis. Management recommendations include decreasing sugars and starches in the diet, increasing exercise, monitoring and maintaining hoof health, and restricting energy intake in obese animals [3]. In many cases, abnormal blood insulin concentrations decrease in response to interventions targeting diet and management, but this decrease can be delayed or incomplete. For patients with suboptimal responses to management interventions, pharmacological treatment is warranted to augment management efforts.

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) have been explored as a treatment for ID in horses because drugs in this family are used to manage type 2 diabetes mellitus in humans [9]. Inhibition of sodium-glucose cotransporters in the proximal tubule of the kidney decreases glucose reuptake, causing glucose to be lost in the urine and lowering blood glucose concentrations [10].

Velagliflozin is an SGLT2i developed for veterinary use. Studies with ID-affected ponies have shown that blood insulin concentrations measured during an OST decrease in response to velagliflozin treatment, with no health complications detected after 39 days of velagliflozin treatment [7, 11, 12]. Ponies treated with velagliflozin did not develop laminitis when subjected to a dietary challenge, in contrast with ponies receiving placebo treatment [7]. The SGLT2i drugs canagliflozin and ertugliflozin currently are used in veterinary practice to treat ID in horses. Short-term effects have been investigated in ponies and horses with ID [13], pharmacokinetics of canagliflozin and single-dose drug effects have been evaluated in healthy Icelandic horses [14], and clinical reports show blood insulin concentrations decrease in response to treatment with these medications [15–17].

Our primary aim was to investigate the effects of velagliflozin treatment on plasma insulin concentration in a population of client-owned horses with naturally occurring ID to test the hypothesis that velagliflozin significantly decreases insulin concentration. The study utilized a double-blind randomized placebo-controlled trial, followed immediately by an open-label trial to evaluate longer-term effects of drug treatment. Secondary study aims were to investigate the effects of velagliflozin treatment on serum glucose and triglyceride concentrations, as well as body condition score [18] (BCS), cresty neck score [19] (CNS), and the modified Obel lameness score [20] (MOL).

2 | Materials and Methods

2.1 | Animals

Adult (aged > 2 years) horses, ponies, and Miniature horses with histories, clinical signs, and laboratory results consistent with ID were included in the study. Donkeys and mules were excluded from participation, as were pregnant or lactating mares.

2.2 | Study Qualification

Horses with known or suspected ID/EMS were recruited by local primary care veterinarians based on medical histories (including being predisposed to obesity or a history of laminitis), clinical signs (regional adiposity or generalized obesity), and laboratory results (hyperinsulinemia or insulin resistance). Horses were excluded from participation if currently receiving any medications expected to affect insulin status, including SGLT2i, metformin, levothyroxine, and glucocorticoids. Horses with previously diagnosed pituitary pars intermedia dysfunction (PPID) were allowed to participate in the clinical trial and remained on any existing PPID treatment regimen (e.g., pergolide, cyproheptadine) throughout the study period. Horse owners agreed to feed a consistent diet over the study period. Because individual horse dietary needs varied among study subjects, diets were recommended by the primary care veterinarian and no feed types were prohibited. Horse owners were asked to maintain a consistent exercise level throughout the duration of the study, as recommended by the primary care veterinarian. Horses undergoing intentional weight loss were excluded from study participation. All horses were maintained in preexisting housing environments for the duration of the study.

All horses being considered for study enrollment underwent a low-dose (0.15 mL/kg body weight) oral sugar test (OST) [21] to confirm a diagnosis of ID, which was performed up to 30 days before study enrollment. All horses were evaluated using this method regardless of whether prior or alternative testing had historically confirmed ID. Only horses with a plasma insulin concentration ≥ 75 μIU/mL at 60 or 90 min during the confirmatory OST were enrolled in the clinical trial.

2.3 | Experimental Design

A 20-week randomized, double-blind, placebo-controlled, parallel-design trial immediately followed by a 20-week open-label trial with delayed start design was performed. Owners, veterinarians, and investigators remained blinded to treatment group allocations. For Study Period 1 (SP1; Weeks 0–20), horses were randomly assigned to Group A (receiving 0.3 mg/kg velagliflozin oral solution PO once per day) or Group B (receiving volume-equivalent placebo PO once per day) using a permuted block design. Owners received labeled bottles of study medication with no indication of whether the bottles contained velagliflozin or placebo. The placebo medication consisted of an oral solution identical to the velagliflozin solution in all of its excipients. For Study Period 2 (SP2; Weeks 20–40), all horses received velagliflozin (0.3 mg/kg velagliflozin PO once per day). At Week 20, all owners started new bottles of study medication, and Group B horses were

immediately switched from placebo to velagliflozin. Treatment group allocation was not revealed to owners or veterinarians until the subjects' participation in the study had concluded. Study visits were performed 11 times during the 40-week clinical trial (at 0, 2, 4, 8, 12, 20, 22, 24, 28, 32, and 40 weeks). Primary care veterinarians obtained owner consent, conducted OST, collected blood samples, performed physical examinations and modified Obel lameness (MOL) evaluations, submitted samples to the laboratory, and sent completed study forms to the investigators. Horses were immediately withdrawn from the study if serum triglyceride concentrations exceeded 1000 mg/dL. In these subjects, serum triglyceride concentration was measured 1–3 weeks after discontinuation of velagliflozin treatment to confirm that serum triglyceride concentrations were normalizing.

2.4 | Study Outcomes

The primary study outcome measure was plasma insulin concentration. Secondary outcomes of serum glucose and triglyceride concentrations, and clinical assessments of BCS, CNS, and MOL score also were explored in the study.

2.5 | Study Visits

A standardized medical history was obtained for each horse at the time of enrollment, including diet, current and prior medications, and history of laminitis, ID, PPID, or any other major medical history (Supplement S1). Study visits were scheduled to occur at a consistent time for each subject throughout the study period (between 0700 and 1000 h for most horses, except where schedules did not permit morning visits) to minimize variability. Blood samples for resting plasma insulin concentration and serum biochemistry were collected in a forage-fed state, with horses having access to only hay or low non-structural carbohydrate pellets. If any grain was fed as part of the subject's routine diet, it was withheld until sample collection was completed and the study visit had concluded. At each study visit, blood samples were collected, and a physical examination and lameness evaluation were performed. Physical examinations were recorded on a standardized data collection sheet, and laminitis was scored out of 12 using the MOL score [20]. Body weight was estimated using a weight tape or heart girth measurement calculation and was used to determine study drug dosing. Body condition was scored out of 9 using the Henneke scale, and cresty neck was scored out of 5 using a previously described scale [18, 19]. The scoring rubrics were provided to primary care veterinarians for reference. Horse owners were instructed to contact primary care veterinarians directly for any medical concerns or emergencies. Primary care veterinarians directly contacted a study investigator (KT) within 24 h of development of any medical concern to discuss case management and ongoing participation in the clinical trial. All study visits were performed between April 2021 and March 2022.

2.6 | Laboratory Sample Collection, Submission, and Analysis

Laboratory samples collected included EDTA blood and whole blood (without any coagulant or separator). Samples were minimally processed by primary care veterinarians or clinical staff to separate EDTA plasma for measurement of insulin concentration and serum for a standard biochemistry panel (including glucose, triglyceride, and creatinine concentrations, as well as sorbitol dehydrogenase, glutamate dehydrogenase, and gamma-glutamyl transferase activities). Samples were shipped overnight (chilled or frozen) to the Animal Health Diagnostic Center at Cornell University (Ithaca, NY) for measurement of requested analytes (Table 1). Insulin concentration was measured by radioimmunoassay (Millipore human-specific insulin kit; EMD Millipore Corporation, St. Louis, Missouri, USA), and serum biochemistry was performed using wet chemistry (Cobas; Roche Diagnostics, Indianapolis, IN, USA).

2.7 | Statistical Analysis

Summary data are reported descriptively as median (interquartile range) unless otherwise described. Analysis of resting insulin and glucose concentrations was performed using generalized linear mixed effects modeling (GLMM) using treatment, sampling week, and the interaction term "treatment by sampling week" as fixed effects and practice site as a random effect. For these analyses, the GLIMMIX procedure in the Statistical Analysis Software (SAS) program was used to account for potential correlation and non-constant variability of the data. For each variable, the baseline measurement (Day 0 value) was included as a covariate. Unstructured covariance was selected to account for the non-even sampling time points. Using data from SP1, GLMM analysis was performed comparing Group A (receiving velagliflozin) to Group B (receiving placebo). Additional analysis of resting insulin, glucose, and triglyceride concentrations was performed using Mann–Whitney *U* testing to compare medians between treatment groups within each study period and between study periods within each treatment group. For BCS, CNS, and MOL, Mann–Whitney *U* testing was used to analyze the change in score for each study period (SP1: Week 20 result compared to Day 0 baseline; SP2: Week 40 result compared to Week 20 baseline).

Except where otherwise indicated, the analyses used all available data. For all analyses, a two-sided 0.05 level of significance was used. Analyses were conducted using SAS Enterprise Guide software, Versions 8.3 and 9.4 (SAS Institute Inc., Cary, NC) and GraphPad Prism, Version 10.0.3 (Graphpad Software LLC, Boston, MA).

3 | Results

3.1 | Study Population

Thirty-seven horses (21 mares and 16 geldings) were included in the study. Twenty-three primary care veterinarians participated in study activities, with each veterinarian enrolling 1 to 4 horses. The median age of study subjects was 21 years (range, 3–29 years) with multiple breeds represented. Thirty-three horses (89%) had a prior history of laminitis (17 in Group A, 16 in Group B), and 22 (61%) had a prior diagnosis of PPID (11 in Group A, 11 in Group B). The median BCS was 6/9 (range, 4–8) and the median CNS was 3/5 (range, 0–4). At the time of enrollment, serum biochemistry results were normal, and the median resting plasma insulin concentration was 191 (range, 20–877) μIU/mL in the study population.

TABLE 1 | Demographic, baseline (Day 0) clinical characteristics, and selected laboratory results for study subjects in Group A and Group B.

Demographic and clinical characteristics	Group A ($N=18$)	Group B ($N=19$)
Age (years)	21 (18–23)	20.5 (18–24) ($N=18$)
Female	11 (61%)	10 (53%)
Horse type		
Horse	10 (56%)	8 (42%)
Miniature horse	5 (28%)	4 (21%)
Pony	3 (17%)	7 (37%)
Laminitis	16 (89%)	17 (90%)
PPID	11 (61%)	11 (61%) ($N=18$)
Body weight (kg)	425 (171–457)	278 (186–416) ($N=18$)
Body condition score	6 (5–7)	6 (5–6) ($N=18$)
Cresty neck score	3 (2–3)	3 (2–3) ($N=18$)
Modified Obel lameness score	2.5 (0–5)	2 (0–4)
Resting insulin (μIU/mL)	188 (127–464)	193 (82–345)
Glucose (mg/dL)	107 (90–129)	108 (90–116)
Triglycerides (mg/dL)	74 (55–90)	81 (57–107)

Note: Data are reported as median (interquartile range) or number (percentage).

Demographic and clinical characteristics of study participants at the time of enrollment are summarized in Table 1.

3.2 | Study Completion

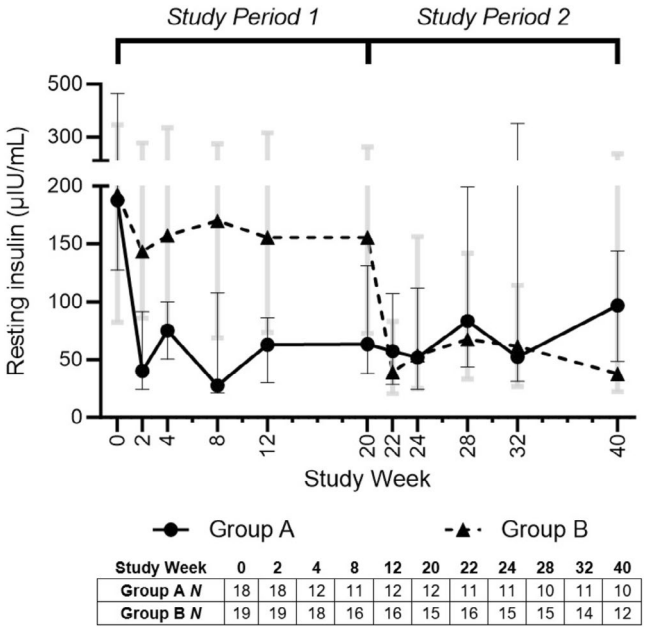
During SP1, 17 of 19 horses in Group B (receiving placebo) completed the 20 weeks of the study and transitioned to SP2. One horse in this group was euthanized because of laminitis (subject with a history of prior laminitis and PPID) and another horse because of suspensory injury (subject with a previously reported abnormality of the suspensory apparatus and no history or laboratory evidence of PPID). Twelve of 18 horses in Group A (receiving velagliflozin) completed SP1 and transitioned to SP2. The six horses that did not complete SP1 developed hypertriglyceridemia (serum triglyceride concentration > 1000 mg/dL), after which the horses were withdrawn from the placebo-control trial. During SP2, two horses in Group B developed hypertriglyceridemia > 1000 mg/dL after beginning treatment with velagliflozin and were withdrawn from the open-label trial. Data from horses removed for hypertriglyceridemia were included in the analysis until the point of removal. Two horses were withdrawn after Week 20 because of owners' scheduling conflicts.

3.3 | Resting Plasma Insulin Concentration

Group A median (interquartile range [IQR]) resting insulin concentration was 71 (33–132) μIU/mL in SP1 and 61 (35–144) μIU/mL in SP2. Group B median (IQR) resting insulin concentration was 157 (82–298) μIU/mL in SP1 and 50 (26–99) μIU/mL in SP2. During SP1, Group A (receiving velagliflozin; median insulin concentration 71 μIU/mL) had significantly ($p<0.0001$) lower insulin concentration than Group B (receiving placebo; median insulin concentration 157 μIU/mL); no significant difference ($p=0.07$) was detected between groups during SP2 (median insulin concentrations 61 and 50 μIU/mL for groups A and B, respectively). Within Group A, no significant difference ($p=0.7$) was found in insulin concentration between the two study periods (median insulin concentrations 71 and 61 μIU/mL for SP1 and SP2, respectively). In contrast, within Group B, insulin concentration was significantly ($p<0.0001$) lower during SP2 (receiving velagliflozin; median insulin concentration 50 μIU/mL) than during SP1 (receiving placebo; median insulin concentration 157 μIU/mL). For SP1, GLMM analysis identified a significant effect of treatment ($p<0.0001$), but not time ($p=0.96$) or the interaction of treatment × time ($p=0.08$), with resting plasma insulin concentration significantly lower in horses receiving velagliflozin compared with horses receiving placebo. Across SP1, the average (95% CI) effect of velagliflozin treatment on insulin concentration was 155 (90–219) μIU/mL. Resting plasma insulin concentrations by study week for both treatment groups are shown in Figure 1.

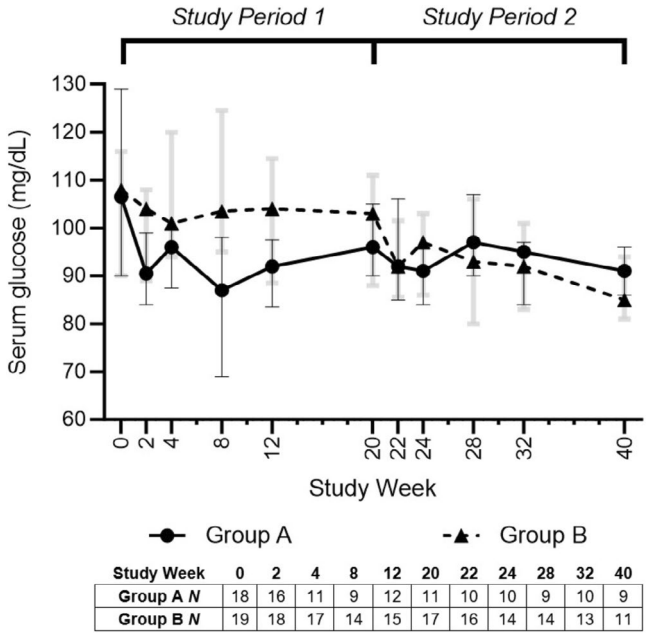
3.4 | Serum Glucose Concentration

Group A median (IQR) serum glucose concentration was 95 (87–105) mg/dL in SP1 and 94 (87–98) μIU/mL in SP2. Group B median (IQR) serum glucose concentration was 104 (92–115) mg/dL in SP1 and 93 (85–103) mg/dL in SP2. During SP1, Group A (receiving velagliflozin) had significantly ($p<0.001$) lower glucose concentration than Group B (receiving placebo); no significant difference ($p=0.6$) was detected between groups during SP2. Within Group A, no significant difference ($p=0.5$) was found in glucose concentration between the two study periods. In contrast, within Group B, glucose concentration was significantly ($p<0.0001$) lower during SP2 (receiving velagliflozin) than during SP1 (receiving placebo). For SP1, GLMM analysis identified a significant effect of



Study Week	0	2	4	8	12	20	22	24	28	32	40
Group A *N*	18	18	12	11	12	12	11	11	10	11	10
Group B *N*	19	19	18	16	16	15	16	15	15	14	12

FIGURE 1 | Median (IQR) resting plasma insulin concentrations by study week in Group A (receiving velagliflozin weeks 0–40) and Group B (receiving placebo weeks 0–20 and velagliflozin weeks 22–40) during Period 1 (Weeks 0–20) and Period 2 (Weeks 22–40). Horses in Group B were switched from placebo to velagliflozin at Week 20 and insulin concentration measured at Week 22 therefore represent the first insulin measurements after initiating velagliflozin administration for Group B horses. Closed circles and solid lines represent Group A and closed triangles and dashed lines represent Group B. The IQR are delineated by thin black lines (Group A) and thick gray lines (Group B).



Study Week	0	2	4	8	12	20	22	24	28	32	40
Group A *N*	18	16	11	9	12	11	10	10	9	10	9
Group B *N*	19	18	17	14	15	17	16	14	14	13	11

FIGURE 2 | Median (IQR) serum glucose concentrations by study week in Group A (receiving velagliflozin weeks 0–40) and Group B (receiving placebo weeks 0–20 and velagliflozin weeks 22–40) during Period 1 (Weeks 0–20) and Period 2 (Weeks 22–40). Horses in Group B were switched from placebo to velagliflozin at Week 20 and glucose concentration measured at Week 22 therefore represent the first glucose measurements after initiating velagliflozin administration for Group B horses. Closed circles and solid lines represent Group A and closed triangles and dashed lines represent Group B. The IQR are delineated by thin black lines (Group A) and thick gray lines (Group B).

treatment ($p=0.001$), but not time ($p=0.56$) or the interaction of treatment × time ($p=0.81$), with serum glucose concentration significantly lower in horses receiving velagliflozin compared with horses receiving placebo. Across SP1, the average (95% CI) effect of velagliflozin treatment on serum glucose concentration was 11 (5–18) mg/dL. Serum glucose concentrations by study week for both treatment groups are shown in Figure 2.

3.5 | Serum Triglyceride Concentration

Group A median (IQR) serum triglyceride concentration was 134 (80–291) mg/dL in SP1 and 77 (61–105) mg/dL in SP2. Group B median (IQR) serum triglyceride concentration was 79 (57–106) mg/dL in SP1 and 181 (97–368) mg/dL in SP2. During SP1, Group A (receiving velagliflozin) had significantly ($p<0.0001$) higher triglyceride concentration than Group B (receiving placebo); during SP2, Group A (continuing treatment with velagliflozin) had significantly ($p<0.0001$) lower triglyceride concentration than Group B (newly started on velagliflozin in SP2). Within Group A, triglyceride concentration was significantly ($p<0.0001$) higher in SP1 than in SP2; within Group B, triglyceride concentration was significantly ($p<0.0001$) higher in SP2 than in SP1. Serum triglyceride concentrations by study week for both treatment groups are shown in Figure 3.

Triglyceride concentration increased quickly after the initiation of treatment with velagliflozin and subsequently decreased throughout the remainder of the active drug treatment period for all horses remaining in the trial, but this finding must be interpreted conservatively because of the withdrawal of horses that exhibited marked hypertriglyceridemia. All horses in Group A exhibited their highest serum triglyceride concentrations in SP1, with 13/18 (72%) occurring at Study Week 2. All horses in Group B exhibited their highest serum triglyceride concentrations in SP2 after starting treatment with velagliflozin, with 12/16 (75%) occurring at Study Week 22 or 24. Eight horses experienced hypertriglyceridemia ≥ 1000 mg/dL. These subjects were withdrawn from the trial, and subsequent triglyceride measurement 1–3 weeks after study withdrawal confirmed decreases in serum triglyceride concentrations in all withdrawn subjects. Development of serum hypertriglyceridemia in study subjects was not accompanied by typical concurrent clinical abnormalities associated with hyperlipemia in horses (inappetence, lethargy).

3.6 | Body Condition Score

At the time of enrollment, horses had a median baseline BCS of 6 (range, 4–8) out of 9 in Group A, and 5 (range, 4–7) out of 9 in Group B. Twenty-six horses had BCS recorded at Day 0, Week 20, and Week 40 (10 in Group A, 16 in Group B). During SP1, 4 horses had an increase in BCS (1 in Group A, 3 in Group B), 14 had a decrease in BCS (8 in Group A, 6 in Group B), and 8 had no change in BCS (1 in Group A, 7 in Group B). In SP1, the median change in BCS from Day 0 baseline was significantly different ($p=0.03$)

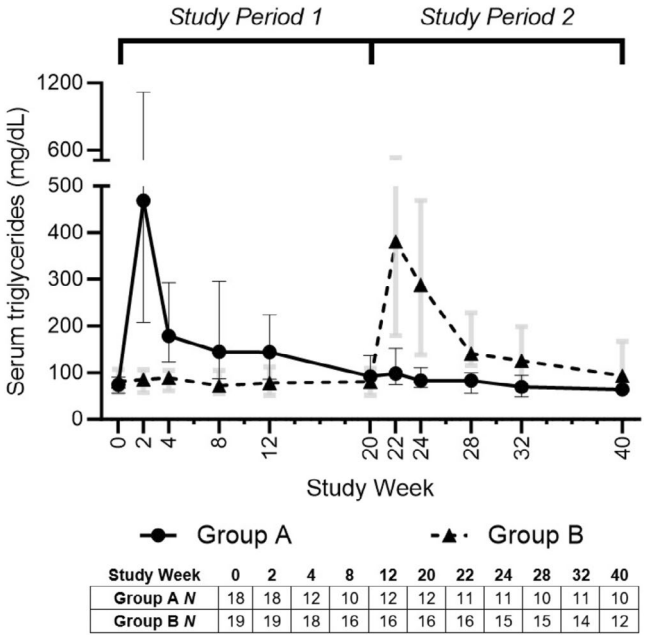


FIGURE 3 | Median (IQR) serum triglyceride concentrations by study week in Group A (receiving velagliflozin weeks 0–40) and Group B (receiving placebo weeks 0–20 and velagliflozin weeks 22–40) during Study Period 1 (Weeks 0–20) and Study Period 2 (Weeks 22–40). Horses in Group B were switched from placebo to velagliflozin at Week 20 and triglyceride concentration measured at Week 22 therefore represent the first triglyceride measurements after initiating velagliflozin administration for Group B horses. Closed circles and solid lines represent Group A and closed triangles and dashed lines represent Group B. The IQR are delineated by thin black lines (Group A) and thick gray lines (Group B).

TABLE 2 | Median (IQR) body condition score (BCS) of study subjects stratified by study week and treatment group.

Study Week	Group (treatment at sampling time)	*N*	Median BCS (IQR)
0	Group A (none)	18	6 (5–6)
	Group B (none)	18	5 (4–6)
20	Group A (velagliflozin)	10	4 (4–5)
	Group B (placebo)	16	5 (5–5)
40	Group A (velagliflozin)	10	5 (4–5)
	Group B (velagliflozin)	16	4 (4–5)

TABLE 3 | Median (IQR) cresty neck score (CNS) of study subjects stratified by study week and treatment group.

Study week	Group (treatment at sampling time)	*N*	Median CNS (IQR)
0	Group A (none)	18	2 (1–3)
	Group B (none)	18	3 (2–3)
20	Group A (velagliflozin)	10	2 (1–2)
	Group B (placebo)	16	3 (2–3)
40	Group A (velagliflozin)	10	1 (1–2)
	Group B (velagliflozin)	16	2 (2–3)

between Group A (receiving velagliflozin; median difference of 1 point lower BCS than baseline; range, 3 points lower to 1 point higher) compared with Group B (receiving placebo; median difference of 0 points lower BCS than baseline; range, 2 points lower to 2 points higher). In SP2 (all horses receiving velagliflozin), the median change in BCS from Week 20 baseline was not significantly different ($p = 0.16$) between horses in Groups A and B. Median BCS is summarized by treatment group and study period in Table 2.

3.7 | Cresty Neck Score

At the time of enrollment, horses had a median baseline CNS of 2 (range, 0–4) out of 5 in Group A, and 3 (range, 0–4) out of 5 in Group B. Twenty-six horses had CNS recorded at Day 0, Week 20, and Week 40 (10 in Group A, 16 in Group B). During SP1, 5 horses had an increase in CNS (1 in Group A, 4 in Group B), 7 had a decrease in CNS (5 in Group A, 2 in Group B), and 14 had no change in CNS (4 in Group A, 10 in Group B). The median change in CNS from baseline was not significantly different between treatment groups during either SP1 ($p = 0.053$) or SP2 ($p = 0.5$). Median CNS is summarized by treatment group and study period in Table 3.

3.8 | Modified Obel Lameness Score

At the time of enrollment, horses had a median baseline MOL score of 2 (range, 0–11) out of 12 in Group A, and 2 (range, 0–7) out of 12 in Group B. Twenty-six horses had MOL scores recorded at Day 0, Week 20, and Week 40 (10 in Group A, 16 in Group B). During SP1, 4 horses had an increase in MOL score (1 in Group A, 3 in Group B), 12 had a decrease in MOL (6 in Group A, 6 in Group B), and 10 had no change in MOL (3 in Group A, 7 in Group B). In SP1, the median change in MOL score from baseline was 2 points lower (range, 10 points lower to 2 points higher) in Group A, and 0 points lower (range, 7 points lower to 7 points higher) in Group B. The median change in MOL from baseline was not significantly different between treatment groups during either SP1 ($p = 0.2$) or SP2 ($p = 0.8$). Median MOL scores are summarized by treatment group and study period in Table 4.

Six horses exhibited hoof pain within the first 10 weeks of the trial requiring additional evaluation and treatment by the primary care veterinarian. Five horses (83%) were receiving the placebo drug at the time hoof pain was observed. The single horse receiving velagliflozin was noted to develop hoof pain on Study Day 5, which corresponded to only a 1-point worsening of the lameness score measurement from Day 0. By the scheduled Study Week 2 visit, this subject's MOL score had decreased, and was 0/12 at the conclusion of the study.

4 | Discussion

Resting serum insulin concentrations decreased when horses and ponies with ID were treated with the SGLT2i velagliflozin.

TABLE 4 | Median (IQR) Modified Obel Lameness score (MOL) of study subjects stratified by study week and treatment group.

Study week	Group (treatment at sampling time)	*N*	Median MOL (IQR)
0	Group A (none)	18	2 (2–5)
	Group B (none)	18	2 (0–3)
20	Group A (velagliflozin)	10	0 (0–1)
	Group B (placebo)	16	0 (0–2)
40	Group A (velagliflozin)	10	0 (0–0)
	Group B (velagliflozin)	16	0 (0–0)

During SP1, horses treated with velagliflozin had significantly lower insulin and glucose concentrations than horses treated with placebo, and for Group B horses, insulin and glucose concentrations were significantly lower during treatment with velagliflozin (SP2) than during placebo treatment (SP1). Hypertriglyceridemia developed rapidly after the initiation of treatment with velagliflozin. The magnitude of this response varied among individual animals, and this laboratory finding was not accompanied by clinical signs of hyperlipemia in horses, such as inappetence or lethargy. Velagliflozin treatment was associated with a significant decrease in BCS during SP1, but significant differences in CNS and MOL between treatment groups were not detected. The main limitations of our study are an overall modest sample size and protocol-mandated withdrawal of patients exhibiting marked hypertriglyceridemia, both of which limit the extent of analyses that can be performed on the study data.

The decreases in blood glucose and insulin concentrations in horses receiving velagliflozin were anticipated given the mechanism of action of SGLT2i drugs, which act to increase urine glucose output in humans. It was assumed that velagliflozin-treated horses experienced glucosuria, but urine glucose excretion was not measured in our study for logistical reasons. Effects of velagliflozin on blood insulin concentrations were first reported when the drug was administered to ponies with ID [7, 11, 12], and our study builds on that work as the first placebo-controlled clinical trial with velagliflozin that includes horses with ID. A related SGLT2i (canagliflozin) was shown to decrease blood insulin concentrations in a single-center placebo-controlled study [13]. Results from our study support prior work suggesting that SGLT2i lower blood insulin concentrations in horses and ponies with ID.

It has been established that hyperinsulinemia induces laminitis in horses, with experimental models as well as epidemiological studies indicating increased risk of laminitis as blood insulin concentrations increase [4–6, 12, 22]. An experimental model utilizing a carbohydrate challenge in ponies with ID found that laminitis developed when plasma insulin concentrations exceeded 195 μU/mL [12]. Consequently, treatments that can effectively mitigate hyperinsulinemia are critically necessary to decrease the risk of clinical laminitis in horses. Mechanisms for the insulin-lowering effects of SGLT2i in horses have not been fully elucidated. These drugs decrease glucose uptake in the renal tubules, and with increased glucosuria, less insulin secretion is likely required to maintain euglycemia. Horses with ID have been shown to exhibit higher beta cell responsiveness to orally administered sugars than healthy horses, and canagliflozin decreases the level of responsiveness in horses with ID [13]. In humans, treatment with the SGLT2i tofogliflozin also is associated with increased hepatic insulin clearance, but similar effects have not yet been established in equids [23]. Further evaluation of the mechanisms by which SGLT2i effect decreases in insulin concentration is warranted.

Serum triglyceride concentrations significantly increased in horses receiving velagliflozin when treatment was initiated, and individual horses differed in the magnitude of hypertriglyceridemia. Interpretation of the changes in triglyceride concentration in our study must be approached judiciously because of the fact that horses with marked hypertriglyceridemia were withdrawn from further participation in the trial, and thus our observations are specific to horses remaining in the trial with limited generalizability. Most horses developed only moderate hypertriglyceridemia, and concentrations decreased over time. Horses that developed hypertriglyceridemia > 1000 mg/dL experienced decreases in triglyceride concentrations within 1–3 weeks after discontinuation of the study drug. Hypertriglyceridemia has been reported previously in response to SGLT2i treatment in horses [16, 24]. Lipid mobilization may be induced by SGLT2i by stimulation of hormone-sensitive lipase, accelerating lipolysis within adipose tissues and saturation of hepatic beta oxidative pathways [25] or by stimulation of lipogenesis. Given interspecies differences in lipid metabolism [26, 27], mechanisms may differ in horses and warrant further investigation. Plasma non-esterified fatty acid, very low-density lipoprotein, and high-density lipoprotein-cholesterol concentrations are increased in horses with EMS compared with normal horses [28]. Thus, treatment with SGLT2i may have differential effects on lipid metabolism in horses with ID compared with unaffected horses. The observation that the severity of hypertriglyceridemia decreased over time in horses remaining in the trial, coupled with the lack of clinical signs exhibited even with marked hypertriglyceridemia, may suggest evaluation of more permissive thresholds of magnitude and duration of SGLT2i-associated hypertriglyceridemia considered tolerable in future studies.

Over the course of the study, BCS and CNS decreased in all horses. The observed significant decrease in BCS in Group A horses during SP1 likely was attributable to loss of energy from the body as glucose was eliminated in the urine [29]. Decreased feed intake was unlikely to contribute to weight loss because inappetence was not reported during the clinical trial, and owners were instructed to maintain the horses' typical diet throughout the study period. Increased voluntary exercise may have contributed to condition improvement. Modified Obel lameness scores improved in all horses, and with better hoof comfort, activity may have increased, but this possibility was not directly evaluated. Although BCS and CNS are subjective measures, these assessments were performed by the subjects' primary care veterinarians using established rubrics to maximize consistency among evaluations. The improvements in insulin concentration and clinical variables in all study horses independent of treatment group could reflect improved owner compliance with recommended dietary and management optimization. A cross-over

study design with Group A horses switching to placebo treatment would have been ideal to evaluate placebo effects, but was not pursued because horse owners expressed concerns about discontinuing a medication with potential clinical benefit at the midpoint of the study; therefore, doing so was considered likely to suppress owner willingness to enroll in the clinical trial.

Our clinical trial recruited client-owned horses and ponies that remained under the care of owners and primary care veterinarians in home environments for the duration of the study. Stress is known to increase blood glucose and insulin concentrations in horses [30], and this study design may have minimized stress in subjects by allowing horses to remain in familiar environments. However, horses were maintained in less controlled environments, and study visits may have caused stress because of the disruption of normal routines. This inconsistency limits nuanced interrogation of the effects of velagliflozin treatment on insulin concentrations, but randomization decreased the risk of bias between the treatment groups. Maintaining usual routines for study horses also was considered to better mirror real-world clinical practice. Requiring adherence to a strict diet and exercise regimen could have established a more consistent baseline across which to compare treatment effects in the study population. However, the extent to which a standardized diet and exercise regimen differed from a subject's usual routine or could be limited by disease comorbidities (such as laminitis) could differentially affect study participants and confound results, and the requirement of a uniform diet and exercise program was considered likely to curtail owner willingness to participate in the trial. All horses in the study underwent insulin testing under forage-fed conditions to increase consistency across the study population, but doing so did not allow for the identification of maximal insulin concentrations using dynamic testing (OST or a dietary glycemic challenge). This approach was intentionally avoided to minimize time demands on the primary care veterinarians, decrease the number of blood collections for each subject, and minimize owner concerns about repeatedly inducing maximal insulin response in the study horses. Another limitation was that horses receiving velagliflozin during SP1 continued receiving the same treatment in SP2 instead of being switched to the placebo. This decision was made to encourage study enrollment and to investigate the longer-term use of velagliflozin for managing ID in horses, mirroring typical clinical practice.

Horses were selected for enrollment in the study using an insulin concentration cutoff of > 75 μU/mL at 60 or 90 min during low-dose OST. The Equine Endocrinology Group recommendations suggest lower thresholds to diagnose ID in clinical patients. However, a higher OST insulin concentration was selected as an inclusion criterion in our study with the goal of enrolling horses with moderate to severe ID because insulin concentrations can provide a measure of laminitis risk [8, 31]. Evaluation of tissue insulin sensitivity could have provided further characterization of ID in horses in our study, but was not considered for this trial because many owners of horses with a history of laminitis express concern about the perceived risk of administering insulin as a part of diagnostic testing.

Study strengths included enrollment of a population consisting of a mix of horses, ponies, and Miniature horses with ID for participation in the trial. The majority (89%) of subjects had histories of laminitis, which is consistent with laminitis being the most serious sequela of ID in clinical patients and reflects the population in which SGLT2i treatment likely would be utilized. Our study was conducted in a manner providing researchers access to patients with naturally occurring clinical disease, while retaining the primary care veterinary team in patient care and maintaining horse owner involvement during study participation.

In conclusion, blood insulin and glucose concentrations as well as BCS decreased in horses and ponies with ID treated with velagliflozin. Transient hypertriglyceridemia developed in horses receiving velagliflozin when the drug was first administered, and the magnitude of this response varied according to the individual animal, suggesting that triglyceride concentrations should be monitored in horses treated with SGLT2i. Our study supports the utility of SGLT2i treatment in horses with ID to decrease resting insulin and glucose concentrations as a component of clinical case management.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the Tufts University Institutional Animal Care and Use Committee (002-20). Authors declare human ethics approval was not needed.

Conflicts of Interest

Nicholas Frank consults for, and Rebecca Voth, Rebecca Klee, Tobias. Warnken, and Victor Chukwi are employed by Boehringer Ingelheim, which provided funds to support this study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supplement S1** Physical Exam Form—Field efficacy of orally administered velagliflozin in the management of insulin dysregulation/hyperinsulinemia in horses. **Supplement S2** Basal Insulin—Individual horse insulin concentrations.